

BEE 208 — Question 2: Cable Temperature Warning System

Control Statement: if...else | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variable temp (double)

Step 3: Prompt user to enter cable temperature in degrees Celsius

Step 4: Read temp

Step 5: If temp > 70, display 'Cable overheating detected.'

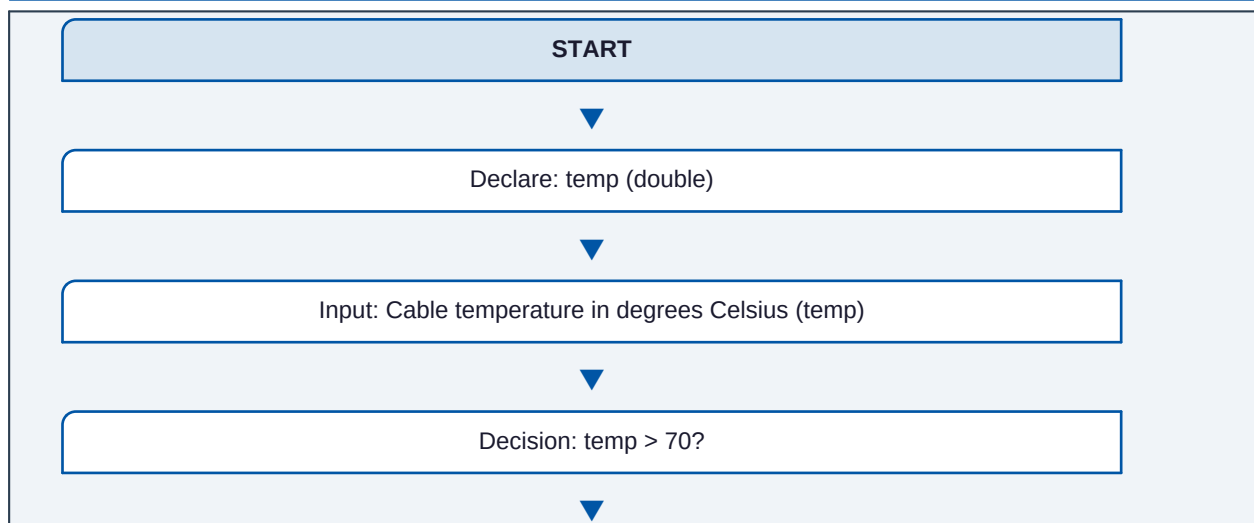
Step 6: Else display 'Cable temperature is within safe range.'

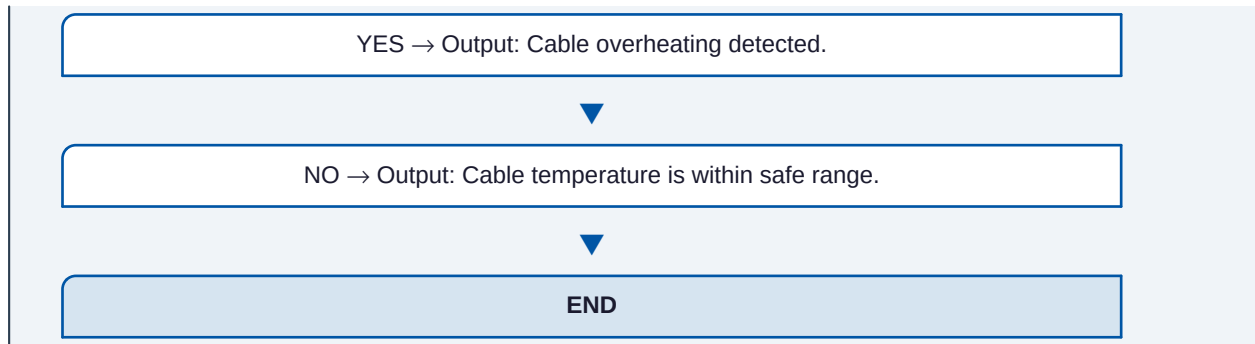
Step 7: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE temp AS DOUBLE  
  
OUTPUT 'Enter Cable Temperature (deg C): '  
  
INPUT temp  
  
IF temp > 70 THEN  
  
    OUTPUT 'Cable overheating detected.'  
  
ELSE  
  
    OUTPUT 'Cable temperature is within safe range.'  
  
END IF  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	temp = 85 deg C	Expected Output	Cable overheating detected.
Input	temp = 55 deg C	Expected Output	Cable temperature is within safe range.